Public Mobile Networks (JPM) — Visual CS Study Script

1. Network Model, Mobility & RF Spectrum (Lectures 1 & 2)

Goal of a Public Mobile Network: Seamless voice, data, and multimedia communication for stationary and moving users without spatial or temporal constraints using a **single address (phone number)**. The architecture splits into a fixed **Core Network** (routing, switching, and databases) and a cellular **Radio Access Network (RAN)**.

- **Terminal Mobility:** A wireless portable device moves across radio coverage areas while keeping active TCP/voice sessions alive. Requires network intelligence to track real-time location and perform seamless connection switching (*handover*).
- **Personal Mobility:** Decoupling the human user from hardware. Introduced via the **SIM card** — your identity and cryptographic keys move into any terminal (wired or wireless) so the network addresses the *person*, not the physical device.
- **Service Mobility:** Your subscribed services (calls, SMS, data) follow you while moving and across foreign operator networks (**Roaming**), with the network automatically selecting the optimal access.

> **WHY DID WE INTRODUCE THE CELLULAR CONCEPT INSTEAD OF ONE GIANT TRANSMITTER? (THE CORE PROBLEM)**
>
> Radio frequency (RF) spectrum (**450 MHz – 3.6 GHz** for mobile networks, wavelength $\lambda = c / f$) is a strictly finite physical resource regulated by ITU-R and national concessions (only narrow bands like **DECT** 1880–1900 MHz and **ISM / Wi-Fi / Bluetooth** 2400–2500 MHz are unlicensed). If we covered an entire city with **one massive transmitter** using 112 channels, only 112 people could talk simultaneously across the whole city! Moreover, a phone would need a huge battery and dangerous transmit power to reach 30 km back to the tower. **The Solution (Cellular Concept):** Divide geography into many small **cells** (idealized as hexagons to tile space without gaps). We can then **reuse the exact same frequencies (Frequency Reuse)** in other cells separated by a safe distance D!

2. Cellular Concept: Cell Cluster (k-Factor), Location Area & Capacity (Lecture 2)

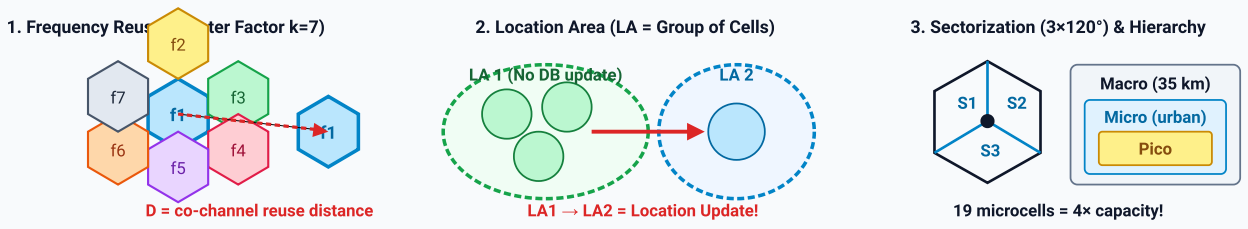


Figure 1: Key radio network scaling mechanisms — Cell Cluster (k-factor), Location Area (LA), and Antenna Sectorization (120°/60°)

- **Cell Cluster (k-factor):** A group of k adjacent cells (in GSM $k = 3, 4, 7, 9, 12$) across which all available operator frequencies are allocated once. A smaller k-factor means a shorter co-channel reuse distance $D \rightarrow$ **higher system capacity, but higher co-channel interference (worse Signal-to-Interference ratio S/I; at cell edges, Signal-to-Noise ratio S/N [or $S/\check{S}$] also degrades)**. *Capacity Calculation (Lecture 2):* With 174 carriers in E-GSM 900 and cluster size $k = 12$, each cell gets $174 / 12 \approx 14$ frequencies. Since each frequency carries 8 TDMA time slots, 1 macrocell (radius r) supports $14 \times 8 = 112$ simultaneous calls. Replacing that single macrocell with **19 microcells** (each using only 3 frequencies out of 9 total) yields $19 \times 3 \times 8 = 456$ simultaneous calls at much lower transmit power!
- **Why did we introduce the Location Area (LA)?** Imagine if 1 million idle phones updated the central database every time a user walked 200 meters into a new urban microcell — the network would suffer a *signaling overload*! Therefore, multiple cells under one MSC are grouped into a **Location Area (LA)**. An idle phone moves between cells inside the same LA **completely silently**, triggering a *Location Update* only when it reads a new **LAI** on the broadcast channel (crossing into a new LA).
- **How do we increase network capacity?** (1) **Cell splitting** (Macro up to 35 km → Micro 100–1000 m → Pico 10–100 m → Nano 1–10 m), (2) **Sectorized antennas** (120° or 60° directional segments, or corner-excited cells where 1 BS covers 3 cells with 120° beams), (3) **Transmit power control**, (4) **Discontinuous Transmission (DTX)**, (5) **Frequency hopping (FH)**, and (6) **Dynamic channel allocation**.

> **PHYSICS OF RADIO WAVE PROPAGATION — WHY AREN'T REAL CELLS NEAT HEXAGONS?**
>
> Radio signals attenuate with distance (fastest in dense urban areas, slowest in rural terrain) and suffer from **reflection** (bouncing off buildings), **diffraction** (bending over sharp obstacles), **scattering and attenuation** (foliage/clouds), and **radio shadows** behind hills. Multiple reflected rays arriving out of phase cause **multipath fading** — severe drops in signal strength over distances of just a few centimeters.

3. GSM Network Architecture: Subsystems, Nodes & Databases (Lectures 1 & 2)

GSM divides the system into the **User Terminal (MS)**, the **Access Network (BSS — Base Station Subsystem)**, and the **Core Network (NSS — Network Switching Subsystem)**:

Node / DB	Full Name (Subsystem)	CS / System Role	Functions & Division of Labor (Lectures 1 & 2)
MS `MT + TE + SIM`	Mobile Station (User Terminal)	Mobile Device + Smart Card Processor	Includes **MT (Mobile Terminal)** (handles radio transmission, speech & channel coding/decoding) and **TE (Terminal Equipment)** — together identified by **IMEI** — plus the **SIM** processor card (stores **IMSI**, auth key **K_i**, PIN, PUK).

Node / DB	Full Name (Subsystem)	CS / System Role	Functions & Division of Labor (Lectures 1 & 2)
BTS `BSS`	Base Transceiver Station	Radio Transceiver Node (1 Cell)	Covers 1 cell (uses 1 to 16 RF frequencies). Performs channel coding/decoding, rate adaptation, encryption/decryption, frequency hopping, paging, and uplink signal measurements.
BSC `BSS`	Base Station Controller	Radio Access Controller	Controls multiple BTSs. Manages radio channels, traffic measurements, frequency hopping, encryption, paging, authentication forwarding, location registration/update, and **handover control**.
MSC `NSS`	Mobile Switching Centre	Core Switching & Call Controller	Controls multiple BSCs. Handles call setup/routing, paging, mobility signaling, location registration, handover switching, SMS, telefax/data services, and **billing/charging data generation**.
GMSC `NSS`	Gateway MSC	External Network Gateway	Interconnection switch linking the GSM network to external networks (**PSTN, ISDN**, other GSM networks). Queries the HLR on incoming calls to route them to the serving MSC.
HLR `NSS DB`	Home Location Register	Permanent Home Subscriber DB	Permanent database of home subscribers: stores **IMSI**, **MSISDN**, authentication key $\mathbf{K_i}$, subscribed services, and the subscriber's **current location (serving VLR)**.
VLR `NSS DB`	Visitor Location Register	Temporary Local Subscriber DB	Associated with every MSC. Temporarily stores subscriber data for both home and roaming users currently inside that MSC's Location Area (deleted once the user leaves the LA).
AUC & EIR `NSS DB`	Authentication Centre & Equipment ID Register	Auth Key DB & Equipment Registry	**AUC (AuC / AC)** holds each subscriber's 128-bit key $\mathbf{K_i}$ for authentication and key generation. **EIR** stores terminal serial numbers (**IMEI**) to verify equipment ownership and detect stolen phones.

4. Identification Numbers & Cryptographic Security in GSM (Lecture 2)

> **WHY DOES GSM SEPARATE MSISDN, IMSI, IMEI, AND TEMPORARY NUMBERS (TMSI, LAI, MSRN, KC)?**
>
> GSM is the first mobile network to **separate subscriber identity (IMSI) from the public dialing number (MSISDN)**. This guarantees identity confidentiality over the radio channel (preventing cloning fraud) and allows modifying phone numbers and routing paths without changing the SIM card:
>
> - **MSISDN (Mobile Station ISDN, max 15 digits = Country code + National mobile number):** The public dialing number assigned with the subscription (e.g., `385 9? xxx xxxx`).
> - **IMSI (International Mobile Subscriber Identity, max 15 digits = MCC [3] + MNC [2] + Subscription Identification [10]):** Unique subscriber ID permanently stored on the SIM card and in the HLR (rarely recycled, only after 6–24 months).
> - **IMEI (International Mobile Equipment Identity = TAC [Type Approval Code] + FAC [Final Assembly Code] + Serial Number):** Uniquely identifies the mobile device hardware; checked via the EIR to block stolen terminals.
> - **Temporary Numbers stored in the MS (TMSI, LAI, MSRN, K_c):** (1) **TMSI (Temporary Mobile Subscriber Identity):** Assigned locally by the VLR so the MS is identified over the air by **LAI + TMSI** — eavesdroppers on radio channels cannot identify the subscriber! (2) **LAI (Location Area Identity):** Broadcast periodically by every BTS on the BCCH so the MS can detect LA changes and trigger a *Location Update*. (3) **MSRN (Mobile Station Roaming Number):** Temporary location-dependent ISDN number assigned via HLR/VLR to route incoming calls from the GMSC to the serving MSC. (4) **K_c (Cipher key):** Temporary 64-bit session key used to encrypt air-interface traffic.

- **GSM Security Services (Access Control/Authentication, Confidentiality/Ciphering, Anonymity) — Algorithms A3, A8, A5:** The 128-bit individual subscriber authentication key **K_i** is stored **only on the SIM and in the AUC (AC)**:
  1. **Authentication (Algorithm A3):** The network sends a 128-bit random number `RAND` to the MS. Both AC and SIM feed the 128-bit `Ki` and 128-bit `RAND` into algorithm **A3** to produce a 32-bit signed response `SRES`. The MSC checks `SRES* =? SRES`.
  2. **Key Generation (Algorithm A8) & Ciphering (Algorithm A5):** Both AC and SIM feed the 128-bit `Ki` and 128-bit `RAND` into algorithm **A8** to generate the 64-bit cipher key `Kc`. Algorithm **A5** in the BSS and MS then encrypts and decrypts voice and signaling data over the radio link.
  3. **2G vs. 3G/4G Security:** In GSM, authentication is unilateral (the MS does not authenticate the BTS, allowing rogue base stations). **3G UMTS** introduces **mutual authentication** (both user and base station are verified), while **4G LTE** introduces hierarchical key derivation from a *Master Secret Key*.

- **Standardized Interfaces & Signaling Protocol Stack (Layers 1, 2, and 3):** Interfaces: `Um` (MS–BTS radio interface), `Abis` (BTS–BSC, 16/64 kbit/s PCM), `A` (BSC–MSC, 64 kbit/s / 2.048 Mbit/s PCM), `B` (MSC–VLR), `C` (MSC–HLR), `D` (HLR–VLR), `E` (MSC–MSC for handover), `F` (MSC–EIR), `G` (VLR–VLR for LA change).
  Protocol Stack: Layer 1 is the physical **TDMA/FDMA** channel (on `Um`) and **PCM** (on `Abis` / `A`). Layer 2 on `Um` is **LAPDm** (*Link Access Protocol D mobile* — fixed 23-octet frame with Address, Control, Length Indicator `LI`, 184-bit Information Field, and Fill; no start/end flags), **LAPD** on `Abis`, and **SS7 (MTP + SCCP)** on `A`. Layer 3 (Message Layer) consists of **RR (Radio Resource Management)** [MS–BSC, bridged via RR' and BTSM on BTS], **MM (Mobility Management)** [MS–MSC], and **CM (Connection / Call Management: CC, SS, SMS)** [MS–MSC, over BSSAP on the A-interface].

> **PHYSICS OF RADIO WAVE PROPAGATION — WHY AREN'T REAL CELLS NEAT HEXAGONS?**
> Radio signals attenuate with distance (fastest in dense urban areas, slowest in rural terrain) and suffer from **reflection** (bouncing off buildings), **diffraction** (bending over sharp obstacles), **scattering** (foliage), and **radio shadows** behind hills. Multiple reflected rays arriving out of phase cause **multipath fading** — severe drops in signal strength over distances of just a few centimeters.

3. GSM Network Architecture: Nodes, Databases & Interfaces (Lectures 1 & 2)

GSM strictly decouples **radio access (BSS)** from **switching and business logic (NSS)**. From a Computer Science perspective, GSM is a distributed system with edge controllers and a two-tier database architecture (master DB + local cache):

Node / DB	Full Name (Subsystem)	CS Analogy	What does it do & why was it introduced? (Functional Split)
MS `ME + SIM`	Mobile Station (User Terminal)	Client Device + Hardware Crypto Token (Smart Card)	Consists of terminal hardware **ME** (identified by **IMEI**) and processor chip card **SIM** (stores **IMSI**, secret key $\mathbf{K_i}$, PIN/PUK). Without a SIM, only emergency calls (112) are allowed.
BTS `BSS`	Base Transceiver Station	Physical Access Point / L1–L2 Radio Modem	Covers 1 cell (1 to 16 RF carriers). Handles physical radio transmission, channel coding, encryption, uplink measurements, and *paging*. Kept intentionally simple so thousands can be deployed cheaply.
BSC `BSS`	Base Station Controller	Edge Controller / Local Radio Orchestrator	Controls multiple BTSs (via $\mathbf{A_{bis}}$ interface). Allocates radio channels, manages frequency hopping, power control, and autonomously executes **intra-BSC handovers** to offload the core network!

Node / DB	Full Name (Subsystem)	CS Analogy	What does it do & why was it introduced? (Functional Split)
MSC `NSS`	Mobile Switching Centre	Core Router / Session Controller	Main switching node (connected to BSCs via **A** interface). Routes calls, coordinates *inter-BSC handovers*, manages mobility, SMS, and generates call billing/charging records.
GMSC `NSS`	Gateway MSC	Ingress/Egress API Gateway / Border Router	Interconnection point to external networks (PSTN, ISDN, other mobile operators). Every incoming external call hits the GMSC first, which queries the HLR to find which MSC currently serves the user.
HLR `NSS DB`	Home Location Register	Master RDBMS (Permanent Subscriber DB)	Permanently stores every home subscriber's profile: **IMSI**, **MSISDN**, allowed services, and a **pointer to the current VLR** where the user is currently located.
VLR `NSS DB`	Visitor Location Register	Local Redis Cache (paired with every MSC)	Temporary database attached to each MSC. When a user (home or roaming) enters an MSC's area, the VLR copies their profile from the HLR so the MSC doesn't query the central HLR on every single call!
AuC & EIR `Security`	Authentication Centre & Equipment ID Register	IAM Auth Server & Hardware Blacklist DB	**AuC** (linked to HLR) stores each user's secret 128-bit key $\mathbf{K_i}$ and generates auth triplets. **EIR** verifies device **IMEI** numbers (white, grey, and black lists to block stolen phones).

4. Identification Numbers & Cryptographic Security in GSM (Lecture 2)

WHY DOES GSM STRICTLY SEPARATE MSISDN, IMSI, TMSI, AND MSRN? (PRIVACY & ROUTING)

In 1G analog networks (NMT), your phone number was transmitted unencrypted over the air — anyone with a radio scanner could intercept it and clone your phone. GSM introduced a major security leap by **decoupling public numbers from internal identities**:

- **MSISDN (Mobile Station ISDN, max 15 digits):** The public phone number dialed by users (e.g., `385 91 xxx xxxx`). Acts purely as a "DNS alias" translated by the HLR into an IMSI. Allows changing phone numbers or porting numbers without replacing the SIM card.
- **IMSI (International Mobile Subscriber Identity, max 15 digits = MCC [3] + MNC [2] + MSIN [10]):** The unique primary key burned onto the SIM and stored in the HLR. **Never transmitted in cleartext over the air** except during initial attachment when no TMSI exists!
- **TMSI (Temporary Mobile Subscriber Identity):** A local, temporary alias assigned by the VLR (and rotated on every Location Update or call). Over the air interface (U_m), the phone is paged and identified strictly by **LAI + TMSI** → an eavesdropper cannot identify who is calling or track a user's movement!

> - **MSRN (Mobile Station Roaming Number):** A temporary ISDN routing number assigned by the HLR/VLR during an incoming call. Why? Because the dialed MSISDN only reveals the user's *home operator*, not *where* they currently are. The MSRN tells the GMSC which exact MSC to route the call to!

- **The GSM Security Triad (Algorithms A3, A8, and A5 — "Challenge-Response"):** The 128-bit secret key K_i is stored **only on the SIM card and in the AuC** and never travels across the network!
  1. **Authentication (Algorithm A3):** The network (AuC/VLR) sends a random 128-bit challenge `RAND` to the phone. The SIM processor computes a 32-bit signed response `SRES = A3(RAND, Ki)` and returns it. If it matches `SRES*` computed by the AuC, the subscriber is authenticated!
  2. **Key Generation (Algorithm A8) & Encryption (Algorithm A5):** Using the exact same `RAND` and `Ki`, both the SIM and AuC independently run **A8** to derive a 64-bit ciphering key `Kc = A8(RAND, Ki)`. Algorithm **A5** then encrypts all voice and signaling traffic over the air between MS and BTS.
  3. **Critical 2G GSM Security Flaw (Fixed in 3G UMTS):** Authentication in GSM is **one-way (unilateral)** — the network authenticates the phone, but the phone never authenticates the base station! This enables *rogue base station (IMSI Catcher / Stingray)* attacks that force encryption off (A5/0). Therefore, **3G UMTS** introduced **mutual authentication**, and 4G LTE introduced hierarchical key derivation (*Master Secret Key*).
- **Standardized Interfaces & Protocol Stack (Layers 1, 2, and 3):** Interfaces: `Um` (air: MS–BTS, uses **LAPDm** on Layer 2 — no flags, fixed 23-octet frame that encodes into 456 bits across 4 bursts), `Abis` (BTS–BSC, LAPD 16/64 kbit/s), `A` (BSC–MSC, uses **SS7**: MTP + SCCP + BSSAP, 64 kbit/s PCM voice), `B` (MSC–VLR), `C` (MSC–HLR), `D` (HLR–VLR), `E` (MSC–MSC for handover), `F` (MSC–EIR), `G` (VLR–VLR on LA change). **Layer 3** signaling splits into: **RR (Radio Resource)** [MS–BSC: channel allocation, TA, power, handover], **MM (Mobility Management)** [MS–MSC: Location Update, IMSI Attach/Detach, authentication], and **CM (Call Management)** [MS–MSC: Call Control CC, SMS, supplementary services].

5. Radio Access (FDD + FDMA + TDMA) & GSM Technical Parameters (Lecture 3)

How can thousands of users communicate simultaneously in both directions without interference? We must solve two problems: **Duplexing** (separating *Uplink* from *Downlink*) and **Multiple Access** (separating different users across space, time, frequency, or code).

- **Frequency Division Duplex (FDD) vs. Time Division Duplex (TDD):** GSM uses **FDD** — Uplink (MS → BTS) uses a lower frequency band (lower path loss to save mobile battery!), while Downlink (BTS → MS) uses a higher band, separated by a fixed **duplex spacing** (45 MHz in GSM 900; 95 MHz in GSM 1800). TDD uses a single frequency band and switches directions rapidly in time.

- **Why does GSM combine FDMA and TDMA?**
  - **E-GSM 900:** Uplink **880–915 MHz**, Downlink **925–960 MHz** (bandwidth 2×35 MHz). Each RF carrier is **200 kHz** wide → **174 carriers** total (classic GSM 900 had 25 MHz = 124 carriers). Modulation is **GMSK** (*Gaussian Minimum Shift Keying*).
  - **GSM 1800 (DCS):** Uplink **1710–1785 MHz**, Downlink **1805–1880 MHz** (2×75 MHz) → **374 carriers**.
  - On each 200 kHz carrier, time is divided into **TDMA frames lasting 4.615 ms**, each containing **8 Time Slots (TS 0–7)** of **577 µs** (156.25 bits). Thus, a **physical channel = (1 carrier frequency, 1 time slot)**. Maximum system capacity is $174 \times 8 = 1392$ channels (E-GSM 900) or $374 \times 8 = 2992$ channels (GSM 1800).

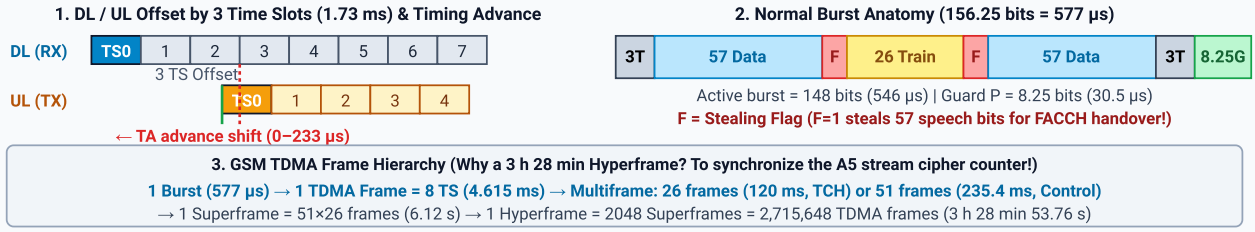


Figure 2: Uplink/Downlink time offset, Normal Burst structure, and TDMA frame hierarchy

6. Critical Physical Layer Concepts: 3-TS Offset, TA & Frequency Hopping (Lecture 3)

WHY DID WE INTRODUCE THESE 3 TECHNIQUES? (HIGH-FREQUENCY EXAM QUESTIONS!)

- **1. Why are Uplink and Downlink TDMA frames shifted by exactly 3 time slots (1.73 ms)?** If a mobile phone (MS) had to receive on TS0 and transmit on TS0 at the exact same instant using a single antenna, its strong transmit (TX) signal would "blind" its sensitive receiver (RX), requiring an expensive, bulky duplex filter. By shifting Uplink by **3 time slots**, the phone first *receives* its burst in TS0, flips a fast switch, and 1.73 ms later *transmits* its burst in TS0 (*Half-Duplex FDD* from the handset's perspective!).
- **2. Why does Timing Advance (TA) exist, and why can't a GSM cell radius exceed 35 km?** In TDMA, 8 different phones share the same frequency in tight 577 µs windows. Radio waves travel at the speed of light ($c = 3 \times 10^8$ m/s), meaning a signal takes 3.33 µs per kilometer (a round-trip delay of 6.67 µs/km). If a phone 35 km away transmitted according to its local clock, its burst would arrive 233 µs late at the BTS and **collide with the next user's time slot**! Therefore, the BTS measures the delay of each phone's *training sequence* and sends back a 6-bit **TA value (0–63, step size 3.7 µs ≈ 550 m)** instructing the phone: *"Transmit your burst TA microseconds EARLIER!"* Because 6 bits (max value 63) can compensate at most **233 µs** of round-trip delay (70 km round trip = 35 km one way), **the maximum GSM cell radius is strictly limited to 35 km**!

- **3. What is the purpose of the 26-bit Training Sequence in the middle of a burst?** As a burst travels through urban streets, multipath reflections smear adjacent bits together (*inter-symbol interference*). Because the receiver knows the exact 26-bit training pattern in advance, it compares the distorted received sequence against the known original, estimates the channel impulse response, and uses an **adaptive equalizer** to reconstruct the surrounding 114 data bits!
- **Slow Frequency Hopping (FH — up to 216 hops/second, 1 hop per 4.615 ms TDMA frame):** TDMA bursts are transmitted across different frequencies in a sequence broadcast on the BCCH. **Why?** Mitigates unwanted multipath signal fading and scatters co-channel interference. **Exception: BCCH and CCCH channels NEVER hop** and always remain on the same frequency (**C0**).

Complete MS – BTS Transceiver Processing Chain (Lecture 3, Slide 20): *Transmit path:* Microphone → A/D conversion (8 kHz, 13 bits) → Segmenting → Speech coding (13 kbit/s) → Channel coding (22.8 kbit/s) → Interleaving → Ciphering → Burst formatting → Modulator → Transmitter. (On the BTS side, 64 kbit/s PCM speech (8 kHz, 8 bits) undergoes D/D conversion to 13 kbit/s). *Receive path:* Receiver → Demodulator → Adaptive equalization → Deciphering → Inverse interleaving → Channel decoding → Speech decoding → D/A conversion → Speaker.

7. Logical Channels & Burst Types in GSM (Lecture 4)

Physical channels (carrier frequency + time slot) carry **logical channels** that separate user payload (**TCH**) from control signaling (**BCH, CCCH, DCCH**). Logical channels are time-multiplexed across 26-frame (120 ms) or 51-frame (235.4 ms) multiframes:

Channel Group	Direction & Multiframe	Logical Channels & Why We Introduced Them
1. Traffic Channels (TCH)	Bidirectional (UL+DL) 26-Frame (120 ms)	**TCH/FS (Full Rate Speech):** Compressed speech at **13 kbit/s** (or data up to 9.6 / 14.4 kbit/s). Occupies 1 full time slot: frames **1–12 and 14–25** carry speech, frame **13** carries **SACCH**, and frame **26 is Idle**. **TCH/HS (Half Rate Speech):** Speech at **6.5 kbit/s** (data 2.4–4.8 kbit/s). Two users share the same time slot on alternating frames, doubling cell voice capacity (frame 26 carries the second user's SACCH!).
2. Broadcast Channels (BCH)	Downlink Only (BTS → MS) 51-Frame (235.4 ms) Always on **C0 TS0**	**FCCH (Frequency Correction):** Sends the *FCB burst* (142 zeros = pure sine wave) so the MS can locate the carrier and lock its local oscillator. **SCH (Synchronisation):** Immediately follows FCCH with an *SB burst* (carries a 64-bit sync sequence with TDMA frame number **RFN** and Base Station Identity Code **BSIC**). **BCCH (Broadcast Control):** Broadcasts cell system parameters: **LAI**, neighboring cell frequencies, hopping rules, and max transmit power.

Channel Group	Direction & Multiframe	Logical Channels & Why We Introduced Them
3. Common Control Channels (CCCH)	Access & Paging 51-Frame (235.4 ms)	**RACH (Random Access — Uplink Only):** Shared *Slotted ALOHA* channel used by the MS to request a dedicated channel (for calls, SMS, or Location Update). Uses the special **Access Burst**! **AGCH (Access Grant — Downlink Only):** BTS reply to a RACH request; assigns a dedicated control channel (SDCCH) and initial **Timing Advance (TA)**. **PCH (Paging Channel — Downlink Only):** Broadcasts paging messages (TMSI/IMSI) to wake up idle phones for incoming calls or SMS.
4. Dedicated Control Channels (DCCH)	Bidirectional (UL+DL) Signaling & Control	**SDCCH (Stand-alone Dedicated):** Temporary signaling channel (often 8 subchannels in 1 TS = *SDCCH/8*) used for authentication, ciphering, call setup (before TCH assignment), and **SMS delivery**! **SACCH (Slow-Associated):** Always paired with a TCH (every 13th frame) or SDCCH. Downlink sends power control and TA updates; Uplink sends MS signal measurements of the 6 strongest neighboring cells. **FACCH (Fast-Associated):** For urgent *handover* commands! Sets the **Stealing Flag (F=1)** to overwrite 20 ms of TCH speech without waiting for frame 13.

8. Five Burst Types, MS Boot Sequence & Channel Combinations (KK1–KK7) (Lecture 4)

- **Why does GSM define 5 distinct burst formats (all 156.25 bits = 577 µs long)?**
  1. **Normal Burst:**
     `3 Tail | 57 Data | 1 F | 26 Training | 1 F | 57 Data | 3 Tail | 8.25 Guard`
     — used for all TCH, BCCH, PCH, AGCH, SDCCH, SACCH, and FACCH channels.
  2. **MS Power-On Boot Sequence & Channel Combinations (KK1–KK7, GSM 05.02):** At power-on, the MS scans RF channels and monitors signal strength → selects the RF channel with the highest level among control channels → waits for **FCCH** (if not detected, picks the next strongest RF channel) → waits for **SCH** (decodes BSIC and RFN) → reads data from **BCCH** and camps on it.
     7 Standard Channel Combinations (KK1–KK7): KK1: `TCH/F + FACCH/F + SACCH/TF` (26-frame multiframe; even TS uses frame 13 for SACCH and 26 idle; odd TS uses frame 13 idle and 26 for SACCH), **KK2:** `TCH/H(0,1) + FACCH/H(0,1) + SACCH/TH(0,1)` (two half-rate users T0/T1 alternate bursts; frame 13 is A0 and frame 26 is A1), **KK3:** `TCH/H(0) + FACCH/H(0) + SACCH/TH(0) + TCH/H(1)`, **KK4:** `BCCH + CCCH + FCCH + SCH` (51-frame multiframe on C0 TS0), **KK5:** `FCCH + SCH + BCCH + CCCH + SDCCH/4 + SACCH/C4` (51-frame multiframe for small cells), **KK6:** `BCCH + CCCH`, **KK7:** `SDCCH/8 + SACCH/C8` (8 dedicated control channels D0–D7 in a single physical channel, 51-frame multiframe).

HYPERFRAME FRAME NUMBERING FORMULA (FN)

Every TDMA frame has a unique Frame Number FN (0 to 2,715,647) derived from 3 counters ($T1$ = superframe index [0–2047], $T2$ = FN mod 26, $T3$ = FN mod 51): `FN = T1 × 26 × 51 + ((T3 − T2) mod 26) × 51 + T3`. This FN feeds into the **A5 cipher** so every 4.615 ms frame is encrypted with a fresh keystream!

9. Speech Coding, Forward Error Correction (FEC) & Interleaving (Lecture 5)

WHY DOESN'T GSM SEND RAW 64 KBIT/S PCM SPEECH, AND WHY SPLIT SPEECH BITS INTO 3 CLASSES?

Landline telephony (ISDN) samples speech at 8 kHz × 8 bits = **64 kbit/s**. Over radio, that wastes too much spectrum. Instead, GSM filters and digitizes speech (8000 samples/s × 13 bits/sample = 104 kbit/s) and feeds it into an **RPE-LTP (RPE-LPC) speech coder** (*Regular Pulse Excitation – Long Term Prediction*). Every **20 ms**, it outputs a block of **260 bits (effective rate 13 kbit/s)**!

Why Unequal Error Protection (Classes Ia, Ib, and II)? Not all 260 bits in a speech block are equally sensitive to transmission errors:

- **Class Ia (50 imperative bits):** Protected by a **3-bit CRC** for error detection (50 + 3 = 53 bits).
- **Class Ib (132 important bits):** Combined with the 53 bits from Class Ia plus **4 Tail bits** (for periodic resetting of the convolutional coder) = **189 bits** fed into a **1:2 convolutional coder** (variables: input $k = 1$, output $n = 2$ [outputs $G0$, $G1$], register length/memory $K = 5$, code rate $R = k/n = 1/2$) → outputs $189 \times 2 = 378$ bits.
- **Class II (78 least important bits):** Appended without error protection ($378 + 78 = 456$ bits every 20 ms → gross rate **22.8 kbit/s**), divided into 4 blocks of 114 bits and interleaved across **8 half-bursts** (interleaving depth in GSM is 4–19).
  Block Coding on Control Channels (Lecture 5, Slide 7): Input $k = 184$ signaling bits + $2t = 40$ parity bits of **Fire code** ($t = 20$) = 224 bits + 4 tail bits = 228 bits → 1:2 convolutional coder = **456 bits**.

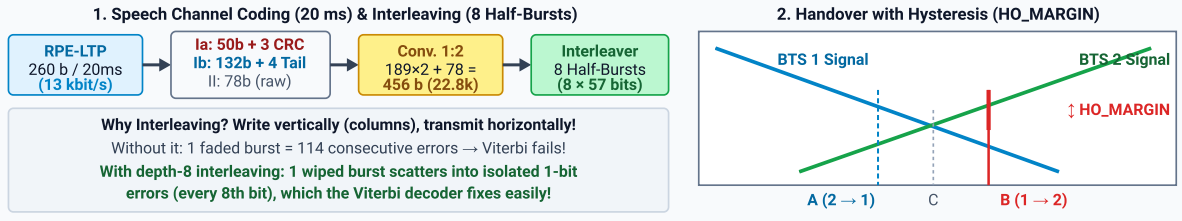


Figure 3: GSM speech channel coding & interleaving pipeline (left) and preventing "ping-pong" handovers via hysteresis HO_MARGIN (right)

10. Radio Link Control: Transmit Power Control (APC) & Handover (Lecture 5)

- **Three Radio Link Measurement Parameters (Lecture 5, Slide 17):**
  1. **RX LEV (Received Signal Level):** Measured on the BCCH channel of the serving BTS and the **6 strongest neighboring BTSs**.
  2. **RX QUAL (Received Signal Quality):** Determined from the Bit Error Ratio (**BER**) before channel coding.
  3. **DISTANCE (Absolute distance between MS and BTS):** Determined using the **Timing Advance (TA)** parameter.
- **Adaptive Power Control (APC):** During call setup, the MS transmits at maximum power, and APC then adjusts transmit power in **2 dB steps** (down to −30 dB below nominal power) to achieve the target BER. **Why?** Reduces co-channel interference, increases spectral efficiency, and saves mobile battery! (*Note:* The BCCH control channel is always transmitted at maximum power).
- **Handover — Channel Sets, Initiation Algorithms & 4 Handover Types:** The MS and network maintain 4 channel lists: **Active set** (currently used), **Candidate set** (good enough for handover), **Neighbor set** (almost good enough), and **Remaining set**. Handover can be *MS-initiated* (in idle mode), *Network-initiated*, or *MS-assisted* (MS reports RSSI, BER, TA, and BS identity to BTS; network decides).
  - **Handover Initiation Algorithms (Lecture 5, Slides 24–27):** (1) *Relative signal strength:* Picks the BTS with the strongest signal (point C in Figure 3, where pathloss $PL(2) < PL(1)$). (2) *Relative signal strength with threshold (*`RXLEV_MIN(1)`, `RXLEV_MIN(2)`*):* Signal level must not drop below minimum threshold; allows handover between points A and B. (3) *Relative signal strength with hysteresis (*`HO_MARGIN(1)`, `HO_MARGIN(2)`*):* Prevents ping-pong handovers! Handover from BTS1 to BTS2 triggers at point B only when BTS1 drops `HO_MARGIN(1)` below BTS2, and from BTS2 to BTS1 at point A via `HO_MARGIN(2)`.
  - **4 Handover Scopes in GSM Architecture:** (1) **Intra-BTS:** Within the same cell (same BTS, different channel to avoid interference). (2) **Intra-BSC:** Between two cells ($BTS_1 \rightarrow BTS_2$) under the same BSC and MSC (BSC activates TCH on BTS_2 → sends `HND_CMD` or `ASS_CMD` on FACCH of BTS_1 → MS sends *handover access burst* to BTS_2 → receives TA on FACCH → sends `Handover Complete` → BSC sends `CLEAR_CMD` to BTS_1). (3) **Intra-MSC:** Between two BSCs under the same MSC (MSC reserves link, target BSC replies with `Handover Ack`, MSC switches voice after TA alignment and sends `CLEAR_CMD`). (4) **Inter-MSC:** Between two MSCs ($MSC_1 \rightarrow MSC_2$). Target MSC_2 asks its VLR to allocate a TMSI and reserves the link to the new BSC, but the **old MSC_1 controls the call until completion** because it holds the subscriber data and call billing/charging information!

WHAT IS DTX (DISCONTINUOUS TRANSMISSION) AND WHY INJECT "COMFORT NOISE"? (LECTURE 6)

In a normal conversation, each person speaks only **40–50% of the time**. **DTX** uses a Voice Activity Detector (**VAD**) to turn off the phone's radio transmitter during silence → cutting battery drain and co-channel interference in half! **The Problem:** When the transmitter shuts off completely, the listener hears "dead air" and thinks the call dropped ("Hello, are you still there?"). **The Fix:** The receiver synthesizes gentle background **Comfort Noise** during silent intervals!

11. MS Operating Modes & Signaling Call Flows (Calls & Location Update) (Lecture 6)

A mobile station operates in one of two states: **Idle Mode** (powered on, no dedicated physical channel, wakes up periodically to listen to its paging group on **PCH** via *Discontinuous Reception — DRX* to save battery, autonomously selects cells) or **Dedicated Mode** (active bidirectional connection on SDCCH or TCH). On exams, all three major signaling procedures (**Outgoing Call MOC**, **Incoming Call MTC**, and **Location Update**) follow the **exact same 4-stage state machine**:

- **Stage 1 — Random Access & SDCCH Assignment (RACH → AGCH):** *(For an Incoming Call MTC, this is preceded by Paging: GMSC queries HLR → HLR fetches MSRN from VLR → call routes to MSC → all BTSs in the user's LA broadcast* `PAG_REQ [TMSI/IMSI]` *on PCH).* The MS sends an Access Burst `CHAN_REQ` on **RACH** (*Slotted ALOHA* — if a collision occurs, it backs off and retries). The BTS measures delay (TA) and forwards `CHAN_RQD` to the BSC. The BSC activates a channel on the BTS (`CHAN_ACT / ACK`) and replies on **AGCH** with `IMM_ASS_CMD` (assigning a dedicated **SDCCH** subchannel and initial **TA**).
- **Stage 2 — Layer 2 Link Establishment (LAPDm) & Security Handshake on SDCCH:** The MS tunes to the assigned SDCCH and sends a LAPDm `SABM` frame piggybacking the initial L3 message (`CM_SERV_REQ` for an outgoing call, `PAG_RSP` for an incoming call, or `LOC_UPD_REQ` for an LA change). The BTS acknowledges L2 with `UA`, and the BSC appends the current `CI + LAC` inside an SCCP `CR` request to the MSC (which replies with `CC`). Next, the MSC/VLR executes: (a) **Authentication:** `AUTH_REQ [RAND]` → MS returns `AUTH_RSP [SRES]`; (b) **Ciphering:** MSC sends `CIPHER_MODE_CMD [$K_c$, A5/X]` (the BTS strips and keeps `$K_c$` locally, forwarding only the algorithm ID `A5/X` to the MS!) → MS replies with encrypted `CIPH_MOD_COM`; (c) **TMSI Reallocation:** `TMSI_REAL_CMD [new TMSI]` → MS confirms with `TMSI_REAL_COM`.
- **Stage 3 — Service Execution (Voice Call Setup or VLR/HLR Database Update):**
  - **For Outgoing/Incoming Calls:** `SETUP` is exchanged (carrying dialed digits or *Bearer Capabilities*) followed by confirmation (`CALL_PROC` for MOC / `CALL_CONF` for MTC). The MSC sends `ASS_REQ`, and the BSC sends `ASS_CMD` instructing the phone to **switch from the signaling SDCCH to a voice TCH channel**! On the new TCH, the MS sends `FACCH/SABM` → `UA` → `ASS_COM` (the temporary SDCCH is immediately freed!). While ringing, `ALERT` is sent; when answered, `CON` and `CON_ACK` start the **active conversation phase on TCH** (with SACCH reports every 480 ms = 4 multiframes).

- ◦ **For Location Update:** The new VLR queries the old VLR over the **G-interface** (`sendIdentification [TMSI]`) to fetch the user's IMSI and auth triplets. Next, the new VLR notifies the HLR via the **D-interface** (`updateLocation`). The HLR sends `cancelLocation [IMSI]` to the old VLR (invalidating the stale cache!) and sends `insertSubscriberData` to the new VLR. The MS receives `LOC_UPD_ACC` .
- **Stage 4 — Connection Release:** Hanging up triggers `DISC` → `REL` → `REL_COM` (L3 Call Control), followed by MSC `CLR_CMD` → `CHAN_REL` (deactivating SACCH, disconnecting LAPDm via DISC/UA, and freeing the radio slot via `RF_CHAN_REL`).

12. Location-Based Services (LBS) & Mobile Positioning (Lecture 7)

WHY IS STANDALONE SATELLITE GPS TOO SLOW FOR MOBILE PHONES, AND WHY DID WE INTRODUCE A-GPS?

LBS applications (emergency **112** calls with automatic caller location, turn-by-turn navigation, fleet tracking, credit card fraud detection matching phone location against POS terminals) require fast, reliable positioning. **The Standalone GPS Problem:** GPS satellites broadcast their navigation message (almanac and precise orbital *ephemeris* data) at a sluggish **50 bit/s**! On a "cold start", downloading this orbital data directly from satellites takes up to **12 minutes (TTFF — Time To First Fix)**, drains the battery, and fails completely in urban canyons if a single packet is lost.

How A-GPS (Assisted GPS) solves this: The mobile operator deploys fixed reference GPS receivers (one covers a ~200 km radius) connected to a Location Server. When your phone needs a fix, the cellular network sends satellite almanac, ephemeris, exact GPS timing, and Doppler corrections at **20–200 kbit/s**! This slashes Time-To-First-Fix from **12 minutes down to ~30 seconds (or a few seconds)** with 10 m accuracy!

Method	Direction	Operating Principle & Mathematical Basis	Accuracy (Urban)	Cost / Limitation
Cell-ID (CI) `+ Sector + TA`	Network / MS	Identifies the serving cell. Adding antenna sector angle (120°) and distance from **TA** (GSM, 550 m step) or **RTT** (UMTS) narrows location to an arc ring.	150 – 500 m (Rural: 1–35 km)	**Zero cost** (no new hardware), but poor accuracy in large macrocells.
AoA (Angle of Arrival)	Uplink (MS → BTS)	At least 2–3 BTSs use phased antenna arrays to measure the **azimuth angle** of the arriving MS signal; intersecting lines pinpoint the user (triangulation).	~100 m	Expensive (requires antenna arrays on every BTS); multipath reflections degrade line-of-sight.

Method	Direction	Operating Principle & Mathematical Basis	Accuracy (Urban)	Cost / Limitation
ToA (Time of Arrival)	Uplink (MS → BTS)	At least 3 BTSs measure absolute signal propagation delay from the MS. The intersection of **3 distance circles** yields the position (trilateration).	100 – 150 m	Requires tight clock synchronization across BTSs and extra **LMU** hardware.
E-OTD (GSM) **OTDOA** (3G/4G)	Downlink (BTS → MS)	The phone measures **time differences** of signals arriving from pairs of BTSs. Each pair defines a **hyperbola** of constant distance difference (hyperbolic trilateration).	50 – 150 m	Good accuracy, but expensive for operators (requires **LMU** units and handset software).
A-GPS / A-GNSS	Satellites + Net	Network supplies assistance data (ephemeris, clock corrections); position is computed either by the phone (*Handset-based*) or by the server from pseudoranges (*Handset-assisted*).	**10 – 20 m** (Rural: 10 m)	Best outdoor accuracy & supports roaming, but **fails indoors**!
Indoor Systems `WLAN / UWB / INS`	Local	Inside buildings (where satellite signals are blocked), positioning relies on picocells (10–50 m), Wi-Fi (802.11), Bluetooth, RFID, UWB (~2 m accuracy), and inertial sensors (gyro/accelerometer).	**1 – 50 m** (UWB: ~2 m)	Short range; requires indoor beacons/mapping or smartphone sensor fusion.

13. Evolution of Cellular Systems: 2.5G (GPRS/EDGE) → 3G (UMTS/HSPA) (Lecture 8)

- **Why were 2G GSM (9.6 / 14.4 kbit/s) and HSCSD unsuitable for the Internet?** GSM is **Circuit-Switched**. **HSCSD** bundled up to 4 time slots via software ($4 \times 14.4 = 57.6$ kbit/s, theoretically 8 TS = 115.2 kbit/s), but a user reading a web page tied up 4 voice channels even while no bytes were flowing!
  - **GPRS (2.5G — General Packet Radio Service):** Introduces **Packet Switching**! Adds two new core nodes: **SGSN** (*Serving GPRS Support Node*) and **GGSN** (*Gateway GSN* — router to the Internet). Over the air, it introduces the shared **PDCH** channel and 4 coding schemes (**CS-1 to CS-4**: block coder adds 40 or 16 Fire code bits + 3, 6, or 12 **USF** [*Uncoded Uplink State Flag*] bits + 4 tail bits; convolutional code rate is $1/2$ for CS-1 [9.05 kbit/s], $2/3$ for CS-2 [13.4 kbit/s], $3/4$ for CS-3 [15.6 kbit/s], while **CS-4 uses no convolutional coding and no tail bits** to reach **21.4 kbit/s per TS** → up to **171.2 kbit/s** across 8 TSs).
  - **EDGE (2.75G — Enhanced Data rates for GSM Evolution):** Replaces GMSK (1 bit/symbol) on the 200 kHz carrier with **8PSK modulation (3 bits/symbol)**. Defines 9 coding/modulation schemes (4 with GMSK: 8.8–17.2 kbit/s; 5 with 8PSK: 22.4–59.2 kbit/s) and new circuit-switched channels **E-TCH/F** (28.8, 32.0, and **43.2 kbit/s** per TS → up to **345.6 kbit/s** across 8 TSs; 474 kbit/s in EDGE and 1.89 Mbit/s in *Evolved EDGE*).

- **3G UMTS (Universal Mobile Telecommunications System — Wideband Code Division Multiple Access (W-CDMA):** UMTS connects both GSM BSS and the new **UTRAN** (*UMTS Terrestrial Radio Access Network*, consisting of **Node B** base stations and **RNC** controllers inside an **RNS** — *Radio Network Subsystem*) to a core network with both a Circuit-Switched domain (MSC/GMSC) and Packet-Switched domain (SGSN/GGSN) plus **IMS** (*IP Multimedia Subsystem*).
  - **Why W-CDMA (5 MHz channel bandwidth) and Spread Spectrum (Lecture 8, Slides 18–24)?** In CDMA, every user device (**UE**) in every cell uses the entire 5 MHz frequency band simultaneously (**Frequency Reuse Factor $k = 1$!**). Before transmission, the narrowband input signal $d(t)$ (bandwidth B_D, power G_D) is multiplied by a fast pseudorandom **PN / Walsh code** $c(t)$ (bandwidth B_C, power G_C) to produce a wideband spread-spectrum signal $d(t)c(t)$ of bandwidth $B_{SS} \approx B_C$ and low power spectral density G_{SS}. At the receiver, the incoming signal with interference $r(t) = d(t)c(t) + i(t)$ is multiplied by the synchronized PN code $c(t)$ to yield $d(t)c(t)c(t) + i(t)c(t)$: a baseband (BB) filter then extracts the narrowband user data while spreading interference $i(t)c(t)$ over B_i!
  - **Rake Receiver (Micro-Diversity) & Soft Handover (Macro-Diversity):** The 5 MHz bandwidth allows a **Rake Receiver** to use multiple sub-receivers tuned to multipath component delays ($\tau_1, \tau_2, \tau_3, \tau_4$) and combine them for a higher signal-to-noise ratio (E_b/N_O). In **Macro-Diversity**, the UE communicates simultaneously with multiple Node Bs using different spreading codes.
  - **Why does "Cell Breathing" occur, and why does UMTS run power control at 1.5 kHz?** Fast power control at **1.5 kHz (1500 Hz)** (vs. 2 Hz in GSM) solves the *near-far problem*. Because capacity in UMTS is **soft**, adding more users increases intra-cell and inter-cell interference noise, so a UE at the cell edge (transmitting at max power) can no longer be heard → **the effective cell range shrinks as the number of users increases (Cell Breathing)!**
  - **3.5G HSPA & HSPA+:** Introduces shared downlink data channel **HS-DSCH** and control channels **HS-SCCH** and **HS-DPCCH**, shorter **TTI**, adaptive modulation (**QPSK, 16-QAM, 64-QAM**), fast retransmission with *soft-combining* of Transport Blocks (TB), and **MIMO** multi-antenna diversity (*SISO, MISO, SIMO, MIMO*).

14. 4G (LTE) and 5G (NR) — Comparative Overview (1G to 5G) (Lecture 8)

- **Why did 4G LTE abandon CDMA in favor of OFDMA (Downlink) + SC-FDMA (Uplink) and an All-IP Core (EPC)?** At wide **20 MHz** bandwidths, CDMA equalization becomes computationally prohibitive. Instead, LTE uses **OFDM**: using **IFFT/FFT** hardware, it splits a wide channel (scalable: 1.4 / 3 / 5 / 10 / 15 / **20 MHz**) into hundreds of narrow, mutually **orthogonal subcarriers spaced 15 kHz apart** (symbol duration $1 / 15$ kHz = 66.7 µs + a 4.7 µs **Cyclic Prefix CP** guard interval that completely eliminates multipath inter-symbol interference!). Resources are scheduled in **PRBs (Physical Resource Blocks)** = 12 subcarriers × 15 kHz = **180 kHz** over 0.5 ms. Adding hundreds of independent OFDM subcarriers creates high power spikes (high —), requiring inefficient, battery-draining amplifiers. transmits symbols serially with constant subcarrier power, ! LTE also eliminates circuit switching entirely — its flat is pure IP with low latency (~4–10 ms), carrying voice via .

- **Why use SC-FDMA on the Uplink instead of pure OFDMA?PAPR***Peak-to-Average Power Ratio***SC-FDMAsaving mobile battery lifeEPC (Evolved Packet Core)VoLTE**
- **5G Networks (New Radio & 5GC):** Built around three ITU-R pillars: (1) **eMBB** (>1 Gbit/s broadband, AR/VR, cloud gaming), (2) **URLLC** (*Ultra-Reliable Low-Latency* — **1 ms** latency for autonomous vehicles, remote surgery, and smart grids), and (3) **mMTC** (massive IoT / smart cities). 5G uses flexible OFDM numerology (subcarrier spacing ***15 / 30 / 60 / 120 / 240*** kHz and up to **256-QAM = 8 bits/symbol**), **Massive MIMO with 3D Beamforming** (steering narrow "pencil beams" directly at individual users to boost throughput and lower environmental RF exposure!), and cloud-native **SDN, NFV, and Network Slicing**. In Croatia (concessions through 2036/2039), 5G uses three key bands: **700 MHz (N28)** for wide rural macro coverage, **3.6 GHz (N78, 100–120 MHz per operator)** for high urban capacity, and **26 GHz (N258, 200–400 MHz)** for multi-gigabit hotspots fed by *optical ring backhaul*.

Generation	Multiple Access	Channel Width	Modulation	Typical Speed	Key Architectural Innovation ("Why was it introduced?")
1G (NMT)	FDMA (Analog)	25 kHz	FM	Voice only	First cellular voice network (1987 in Croatia); unencrypted analog radio.
2G (GSM)	FDMA + TDMA (8 TS)	200 kHz	GMSK (1 b/sym)	9.6 – 14.4 kb/s	Digital speech (13 kb/s), SIM card (IMSI/TMSI), A3/A8/A5 encryption, SMS.
2.5G (GPRS)	FDMA + TDMA (PDCH)	200 kHz	GMSK (CS 1–4)	up to 171.2 kb/s	**Packet switching** in the core (SGSN + GGSN); always-on per-MB billing.
2.75G (EDGE)	FDMA + TDMA	200 kHz	**8PSK (3 b/sym)**	up to 384–474 kb/s	Triples data rate on existing 200 kHz GSM carriers via 8PSK modulation.
3G (UMTS)	**W-CDMA** (Codes)	**5 MHz**	QPSK	144 kb/s – 2 Mb/s	Spread spectrum, **Reuse k=1**, Rake receiver, Soft Handover, fast APC (1.5 kHz).
3.5G (HSPA+)	W-CDMA + Shared Ch.	5 – 10 MHz	16-QAM / 64-QAM	14.4 – 84 Mb/s	Short 2 ms TTI, adaptive modulation, HARQ soft-combining, and **MIMO** antennas.
4G (LTE / LTE-A)	**OFDMA** (DL) **SC-FDMA** (UL)	1.4 to **20 MHz** (PRB = 180 kHz)	QPSK, 16-QAM, 64-QAM (6 b/sym)	100 – 326 Mb/s (LTE-A >1 Gb/s)	**All-IP flat core (EPC)** with no circuit domain, low latency (~4 ms), VoLTE, 8×8 MIMO.
5G (NR / 5GC)	OFDMA / SC-FDMA (Scalable Δf)	20 to **400 MHz** (700M / 3.6G / 26G)	up to **256-QAM** (8 b/symbol)	**>1 – 10 Gbit/s** (1 ms latency)	**Massive MIMO + 3D Beamforming** (pencil beams), SDN/NFV, **Network Slicing**, URLLC.